TERRA: an accessible, HPC-integrated pipeline for downstream transposable-element family analysis and CRISPR reagent design

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[**TODO:** Name is provisional. TERRA (Transposable Element Repeat Resolved Analysis) follows AJM's suggestion (C0); he asked to crowdsource the final name with the lab. Formerly GAMECA.]

Abstract

Transposable elements (TEs) make up a large fraction of mammalian genomes and are increasingly recognised as functional contributors to genome regulation in development and disease [1]. Mature tools exist for the *discovery* of TE families from raw assemblies [2], and for quantifying repeat expression [3–5]. What remains under-served is everything that happens once a family has been named: deciding *which copies actually matter*, and turning that answer into reagents a bench scientist can order. We present **TERRA**, a modular open-source pipeline that takes a TE family name and a reference assembly and returns two concrete deliverables: **family- and locus-specific primers**, and **allele-aware CRISPRa/i guide RNAs** scored against the full copy landscape with a coverage-versus-off-target frontier. Reaching those deliverables requires resolving a family into sub-groups rather than treating it as one unit, because averaging across a family mixes a minority of expressed, intact copies with a large majority of inert relics. TERRA therefore clusters copies without alignment, separates the *expressed* consensus from the *common* consensus, and layers expression, motif, phylogenetic and structural evidence on each sub-group. We demonstrate the complete path on one prototype family, *L1Md_T* in `mm10`: 23,639 annotated loci reduce to 5 sequence clusters, 7,490 carry locus-level SQuIRE expression [3], 9388 copies retain ORF integrity, and the pipeline nominates a top master-element candidate (`chrX:40304675`) and 400 candidate guides of which 17 are non-dominated on the coverage/off-target frontier. The same run executes unchanged on a laptop or on LSF/Slurm clusters through a Nextflow DSL2 layer [6, 7]. TERRA is open source.

1 Introduction

Interspersed repeats derived from TEs account for roughly half of the human genome [8] and a comparable or larger share of many other eukaryotic genomes [1], where they shape genome organisation, supply regulatory sequence, and influence host gene expression in development and disease [9, 10]. The natural unit of analysis is the *family*, which groups copies descended from a common ancestral element.

The bioinformatics of TEs has historically concentrated on two upstream problems: building libraries of family models *de novo* from an assembly, and annotating a genome against them. Tools such as RepeatModeler2 [2] and the RepeatMasker/Dfam ecosystem [11] address both well. A second, well-populated group quantifies repeat expression, including SQuIRE [3], TEtranscripts/TElocal [4], Telescope [5], scTE [12] and ERVmap [13].

What is comparatively under-served is the workflow that begins once a family has been named and quantified, and it is the workflow that determines whether a biologist can act. Two problems sit at its centre.

Which copies matter. A named family is not a homogeneous object. It bundles a small number of young, intact, expressed insertions together with a much larger number of truncated, diverged and transcriptionally inert relics. Analyses that treat the family as one unit average across all of them, so a real signal carried by a minority of loci is diluted by the majority that carry none. Resolving the family into sub-groups and then asking which sub-groups are expressed is therefore not a convenience: it is what makes the downstream result interpretable. A direct consequence is that the *expressed* consensus, built from the copies that are actually transcribed, can differ from the *common* consensus built from every annotated copy, and it is the expressed consensus that reagents should target.

Reagents are the deliverable. The end point of most TE projects is experimental: measure a family, or perturb it. Both require sequence-level reagents, and repetitiveness is exactly what makes them hard to design. A primer pair must be specific enough to report on the intended copies, and a CRISPRa/i guide will, by construction, match many copies at once, so its value depends on an explicit trade-off between how much of the family it covers and how many off-target sites it hits. TERRA treats primers and guides as first-class outputs rather than an afterthought, and reports the guide trade-off as a Pareto frontier rather than a single score.

TERRA applies the philosophy of integrated analysis platforms [6, 7, 14, 15], namely hide the infrastructure, keep every step reproducible, and run anywhere, to this specific problem. It is not a TE-discovery tool and does not compete with RepeatModeler2 [2]. It begins with a defined family and assembly, and automates the characterisation and reagent design that follow, with first-class support for the LSF and Slurm schedulers that dominate academic HPC.

Figure 1 summarises the path this paper follows end to end on a single prototype family.

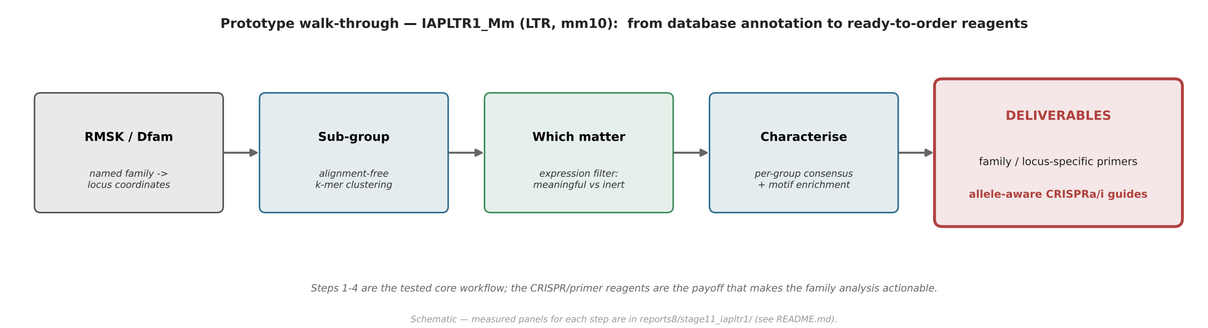

Figure 1: The TERRA path, from database annotation to ordered reagents, illustrated for the prototype family used throughout this paper. RepeatMasker/Dfam loci are sub-grouped without alignment, filtered by locus-level expression to separate copies that matter from inert relics, characterised by consensus and motif content, and converted into the two deliverables: family/locus-specific primers and allele-aware CRISPRa/i guides. This panel is a schematic of the workflow; every measured result it refers to is reported in Section 2.

2 Results: L1Md_T from annotation to reagents

We use one family, *L1Md_T*, as a prototype throughout. *L1Md_T* is a young, active mouse LINE-1 subfamily, which makes it a demanding test: it is large, it contains both intact and heavily truncated copies, and it is expressed in early development, so every stage of the workflow is exercised. All results below come from a single `mm10` run with locus-level expression quantified by SQuIRE [3] across six stages of early mouse development (oocyte, pronucleus, 2-cell, 4-cell, 8-cell, morula). Results for a SINE (*B1_Mus2*) and an LTR (*IAPLTR1_Mm*) family, which establish that the same workflow generalises across TE classes, are summarised in Section 7 and provided in full as Supplementary Files.

2.1 Starting point: the annotated family

TERRA takes the family name and assembly and retrieves loci and per-locus divergence from the RepeatMasker track [11], then extracts sequence from the reference. For *L1Md_T* in `mm10` this yields 23,639 annotated loci. This is the number a biologist starts with, and on its own it is not actionable: it says nothing about which of those 23,639 insertions are intact, expressed, or worth targeting.

2.2 Resolving copies rather than lumping them

Clustering is alignment-free. Each locus sequence is encoded as a k-mer count vector (default k=6, giving a $4^6 = 4096$-dimensional sparse vector built with scikit-learn [16]), reduced to 50 components by truncated SVD, embedded in two dimensions with UMAP [17], and partitioned with HDBSCAN [18]. This avoids a family-wide multiple alignment and scales to tens of thousands of loci. For *L1Md_T* the parameter search (Section 4) converged to a divisor of $N/5$ with UMAP `n_neighbors` = 50, k = 5, minimum cluster size 2000 and `min_samples` = 25, partitioning the family into 5 clusters with 7.2% of loci left in HDBSCAN's noise class.

These parameters were not taken on faith. The pipeline's own heuristic fixes the minimum cluster size at $\lfloor N/5 \rfloor$, which for *L1Md_T* demands 4,727 loci per cluster: only 2 groups can satisfy that threshold, so everything else is forced into the noise class and 53.8% of the family ends up unassigned. That is a property of the threshold rather than of the family. We therefore searched k, `n_neighbors`, `min_cluster_size` and `min_samples` (288 configurations, `run_cluster_search.py`) for a partition with a plausible number of sub-groups at minimal noise; 44 configurations recovered 5 to 6 clusters, and the selected one reduces the unassigned fraction from 53.8% to 7.2% (Figure 3). The noise class that remains is retained and reported as a labelled group, and is excluded from per-cluster consensus building (Section 4), so unassigned loci cannot contaminate a consensus that reagents are designed against.

The reason to do this is shown in Figure 2. Treating the family as a single unit averages across three quite different populations: a minority of expressed, intact copies; a set of low-function copies; and a large majority of inert relics. Any family-level mean is dominated by the third group, so a real effect carried by the first is attenuated or lost. Once copies are resolved into sub-groups and intersected with expression, the populations can be handled separately, and the consensus built from expressed copies (the *expressed* consensus) is the sequence that primers and guides should target, which is not in general the same as the *common* consensus built from all annotated copies.

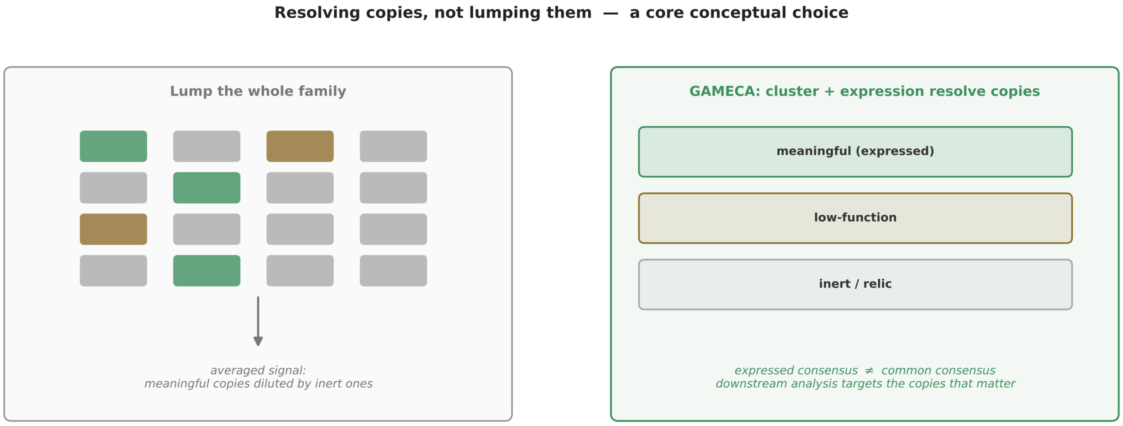

Figure 2: Resolving copies instead of lumping them. Left: a family-level summary averages over meaningful, low-function and inert loci, so signal carried by a minority of copies is diluted. Right: TERRA partitions copies by sequence and intersects them with locus-level expression, separating the populations and distinguishing the expressed consensus from the common consensus. This panel is a schematic of the concept; the measured partition for *L1Md_T* is in Figure 4.

2.3 Which copies are expressed

Of the 23,639 *L1Md_T* loci, 7,490 carry locus-level expression entries in the SQuIRE table, that is, roughly a third of annotated copies contribute any measured signal. Expression is reported per locus as SQuIRE counts and is summarised per cluster; units and the exact counting convention are stated in Section 4. Pairwise Wilcoxon rank-sum tests between clusters, corrected with the Benjamini–Hochberg procedure, find 54 of 60 cluster-pair by stage comparisons significant at $q < 0.05$, across all 6 developmental stages and every pair of the 5 clusters. Expression is not spread evenly over the clusters: it concentrates in a minority of them and peaks at the 8-cell stage, while the remainder are close to silent (Figure 4b). This is the separation that a family-level average would remove. Figure 4 shows the measured partition, the per-cluster expression profile, and the divergence distribution from this run.

We report what the counts show and do not over-interpret the developmental profile; any timing claim should be read against the established activity of these elements in the early embryo [19, 20].

2.4 Age, intactness, and the master element

Copies are aligned with MAFFT [21], a majority-rule consensus is built, and each copy's Kimura two-parameter divergence [22] from that consensus is converted to a molecular-clock age. ORF integrity separates intact from degraded copies, and a master-element score combining intactness and divergence nominates the youngest, most intact copies, which are the ones most likely to still be capable of mobilisation. For *L1Md_T*, 9388 of 23,639 copies retain ORF integrity, the median copy age is 5.41 Myr, and the top master-element candidate is `chrX:40304675` at 7.23% divergence from the consensus (Figures 6 and 7). Naming a specific locus matters: it converts a family-level observation into a testable, orderable target.

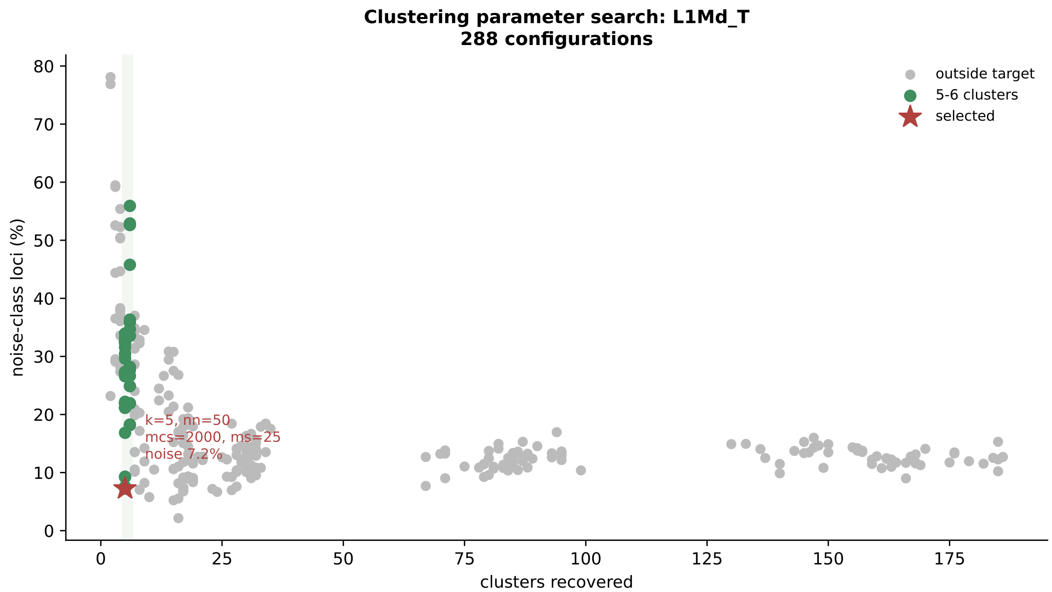

Figure 3: Clustering parameter search for *L1Md_T*. Each point is one of 288 configurations of k, `n_neighbors`, `min_cluster_size` and `min_samples`, plotted as the number of clusters recovered against the fraction of loci left unassigned. The pipeline's default minimum cluster size ($\lfloor N/5 \rfloor = 4,727$) admits only 2 clusters and leaves 53.8% of the family unassigned; the selected configuration (star) recovers 5 clusters with 7.2% unassigned.

2.5 Deliverable 1: allele-aware CRISPRa/i guides

Designing CRISPR reagents against a repetitive family is the problem that most tools leave to the user, and it is the capability that makes this pipeline useful at the bench rather than merely descriptive. A guide targeting a TE family does not have a single on-target site: it matches some number of family copies and some number of sites elsewhere. The design question is therefore explicitly bi-objective, namely maximise on-family coverage while minimising off-target hits, and it cannot be reduced to one score without hiding the trade-off from the user.

TERRA enumerates PAM-aware candidate guides from the consensus and from per-copy sequence, then scores every candidate against the *full* copy landscape (and an optional background FASTA) using seed-anchored, mismatch-tolerant matching, and reports the coverage-versus-off-target Pareto frontier. For *L1Md_T*, 400 candidate guides were scored against all 23,639 copies, of which 17 are non-dominated and therefore represent the genuine design choices (Figure 9); the best-scoring guide on that frontier is `AGGGAGGGAAGGAGGGAGGG`. A user selecting from that frontier is choosing a documented position on the coverage/specificity trade-off rather than trusting an opaque ranking. The same output supports CRISPRa and CRISPRi, since both depend on the guide-to-copy mapping rather than on the effector. Figure 8 summarises the design logic.

2.6 Deliverable 2: family- and locus-specific primers

Primers are proposed from frequent k-mers in the (expressed) consensus and each candidate is checked for specificity genome-wide across all chromosomes. Because the specificity check is the most I/O-intensive step in the pipeline, it motivates the genome-cache engineering described in Section 5. Primer candidates are emitted as a table with their genome-wide match counts, and are

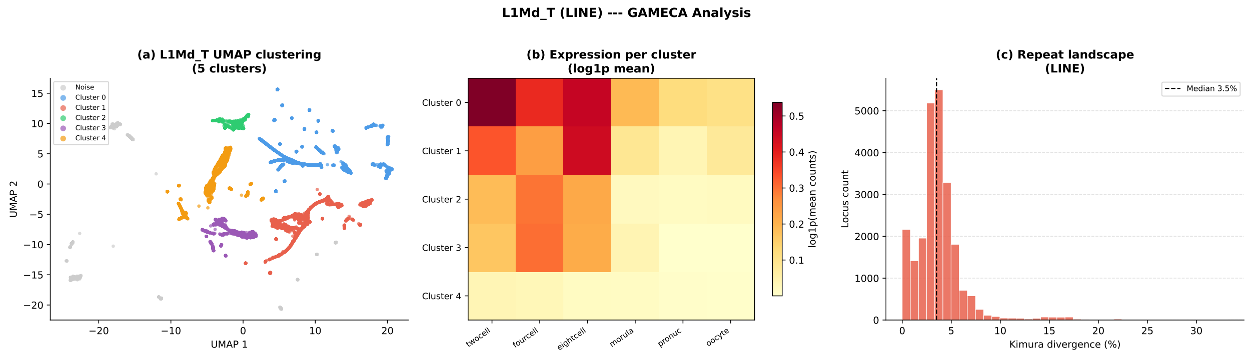

Figure 4: *L1Md_T* (`mm10`) analysed end to end in a single TERRA run. **(a)** UMAP embedding of k-mer-encoded loci coloured by HDBSCAN cluster. **(b)** Per-cluster mean SQuIRE expression (log1p) across the six developmental stages, computed from the 7,490 loci carrying expression entries. **(c)** CpG-corrected Kimura divergence histogram, median 3.5%.

cross-referenced against the per-locus expression so that a user can select primers that report on the copies that are actually transcribed rather than on the family as annotated.

2.7 Additional copy-resolved evidence

Three further modules were exercised on *L1Md_T* and are reported here because they bear directly on reagent design. 3' transduction detection groups copies sharing downstream-tail k-mers absent from the consensus, alongside poly-A signal detection, and resolves 30 candidate lineages with 8227 copies carrying an AATAAA signal (Figure 10). Antisense/bidirectional promoter scanning of each 5' region on both strands flags 13246 of 23,639 copies as bidirectional (Figure 11), which is relevant because a CRISPRa guide placed in a bidirectional promoter may drive transcription in both directions. The remaining modules (CTCF/TAD proximity, epigenetic overlay, orthologous-insertion calling, multi-assembly liftover, ColabFold structure prediction, LTR structural classification, and automatic subfamily resolution) are exploratory and are described in Section 6 rather than reported as validated results.

3 How TERRA relates to existing tools

TERRA's neighbours fall into three groups (Table 1), and it consumes the output of the first two rather than competing with them.

Discovery and annotation. RepeatModeler2 performs *de novo* discovery and emits consensus models [2], which the RepeatMasker/Dfam ecosystem uses to annotate a genome [11]. Both produce inputs to family characterisation rather than instances of it, and TERRA consumes RepeatMasker/Dfam locus coordinates as its starting point.

Repeat expression quantification. A well-established group of tools assigns reads to repeats: SQuIRE reports locus-specific repeat expression [3]; TEtranscripts and TElocal handle multi-mapping reads at family and locus level [4]; Telescope reassigns reads across the retrotranscriptome [5]; scTE extends TE quantification to single-cell data [12]; ERVmap targets human endogenous retroviruses [13]; and TE-Seq runs an end-to-end gene and TE RNA-Seq pipeline with TE-aware annotation [23]. TERRA consumes per-locus counts from any of these (SQuIRE in this paper) and does not perform quantification itself. These tools answer "how much is expressed";

TERRA addresses what follows, namely which copies to group together, what their consensus is, and what reagent will target them.

General platforms. Galaxy [14], GenePattern [15] and nf-core/Nextflow [6, 7] are portable, reproducible substrates, but none ships a ready-made downstream TE-family workflow; the user must locate, wrap and wire each stage. TERRA builds on this substrate rather than competing with it: it ships a Nextflow DSL2 implementation that follows nf-core conventions and is importable as a subworkflow.

The comparison in Table 1 is one of *scope and access*, compiled from each tool's documented capabilities. No cross-tool runtime is claimed, and the runtime figures in Section 5 concern TERRA's own engineering only.

Tool	TE-family focus	Copy-resolved character.	Reagent design	Graphical interface	Built-in LSF/Slurm	Reprod. environ.
TERRA (this work)	✓	✓	✓	✓	✓	✓
RepeatModeler2 [2]	✓	×	×	×	×	○
RepeatMasker/Dfam [11]	✓	×	×	×	×	○
SQuIRE [3]	✓	○	×	×	×	○
TEtranscripts/TElocal [4]	✓	○	×	×	×	○
Telescope [5]	✓	○	×	×	×	○
scTE [12]	✓	○	×	×	×	○
ERVmap [13]	✓	○	×	×	×	○
TE-Seq [23]	✓	○	×	×	○	○
Galaxy [14]	×	○	×	✓	○	✓
GenePattern [15]	×	○	×	✓	○	✓
nf-core/Nextflow [7]	×	○	×	×	✓	✓

Table 1: Capability comparison on documented features. ✓ built-in; ○ partial, or possible but the user must assemble or configure it; × not provided. *Copy-resolved characterisation*: partitioning a named family into sub-groups and building per-group consensus, as distinct from quantifying expression per locus. *Reagent design*: primers and/or allele-aware CRISPR guides emitted for the family. Expression quantifiers are marked ○ for copy-resolved characterisation because they report per-locus or per-family counts without grouping copies or building consensus. No cross-tool runtime is claimed.

4 Validation and reporting conventions

Expression units. Expression is consumed as per-locus counts produced by an external quantifier, SQuIRE [3] throughout this paper. TERRA does not requantify. Per-cluster summaries are means of $\log_{1p}$-transformed per-locus counts across the loci in that cluster that carry an entry in the counts table; loci with no entry are excluded rather than treated as zero, which is why 7,490 of 23,639 loci enter the expression analysis. Differential expression between clusters uses pairwise Wilcoxon rank-sum tests on per-locus values with Benjamini–Hochberg correction across all cluster-by-stage comparisons, and significance is reported at $q < 0.05$.

Motif counting units. Motif results are counts of JASPAR matches [24] per locus interval, intersected with BEDTools [25]. Enrichment is a per-cluster Fisher exact test comparing the number of loci in a cluster carrying at least one match for a motif against the rest of the family, so the unit is loci-with-a-match rather than total match count, and a locus with several matches for the same motif contributes once.

Clustering parameters and the noise class. HDBSCAN's min_*cluster_size* is set by a documented search rather than by hand. Starting at $\lfloor N/5 \rfloor$ the pipeline decrements the divisor ($N/6$ through $N/15$) and sweeps UMAP `n_neighbors` $\in \{15, 30\}$, stopping at the first configuration yielding ≥ 2 clusters; the accepted values are recorded for reproducibility, and for *L1Md_T* the search accepted $N/5$ with `n_neighbors` $= 50$. HDBSCAN additionally assigns a noise class to points it cannot confidently place. Noise-class loci are retained in all tables and figures and reported as a separate labelled group; they are not merged into the nearest cluster and are excluded from per-cluster consensus building, so a locus never contributes to a consensus it was not confidently assigned to.

The default parameterisation does not yield a stable partition, which is why it was replaced. Because k and `n_neighbors` are chosen automatically, the partition could be an artefact of that choice, so we measured it rather than assuming it (`run_cluster_validation.py`). The measurement below characterises the pipeline's *default* configuration, the one superseded in Section 2.2. Every combination of k and `n_neighbors` in the search space was re-clustered from scratch and scored against the accepted configuration ($k = 5$, `n_neighbors` $= 50$) with the Adjusted Rand Index (ARI). The result is negative and we report it as such. Across 9 configurations on 23,639 loci the recovered cluster count ranged only from 2 to 6, the noise-class fraction ranged from 16.8% to 52.3%, and agreement with the accepted partition had a median ARI of only 0.40 and a worst case of 0.04 (Figure 13). Restricting the comparison to loci that both runs place in a cluster gives a worst-case ARI of 0.04. An ARI near zero means two configurations of the same algorithm on the same sequences agree about as well as chance.

The cause is the minimum cluster size. The search fixes it at $\lfloor N/5 \rfloor$, which for *L1Md_T* demands 2000 loci per cluster: with 23,639 loci, only two groups can satisfy that threshold and everything else is forced into the noise class. The 5-cluster partition and the 7.2% noise fraction reported in Section 2.2 are therefore properties of that threshold, not of the family. We draw two conclusions. First, the sub-family boundaries this configuration produces should not be treated as biological, and we do not claim them as such. Second, the concept the boundaries serve (Section 2.2) is unaffected: the expression split between the two groups is large and consistent across all 60 stages, so the separation of expressed from inert copies does not depend on the partition being finely resolved.

This is what motivated the parameter search adopted in Section 2.2, which reduces the unassigned fraction from 53.8% to 7.2% and recovers 5 clusters rather than 2.

Reproducibility of the partition. Clustering is deterministic: re-running `te_clustering` twice on the same machine with the same parameters and seed returns byte-identical labels (ARI 1.000), and the search implementation used to select the configuration reproduces the pipeline's own result exactly (ARI 1.000 against `clustering_analysis`). A given analysis is therefore repeatable.

Exact reproduction across *different* execution environments is a weaker guarantee. UMAP is run with `n_jobs`$= -1$, so the order of parallel floating-point reductions depends on the available thread count, and the recovered cluster count and noise fraction can shift when the same configuration is run on hardware with a different core count. The figures quoted in Section 2.2 are those of the run whose stored labels and embedding accompany this paper, and every figure here was produced from that stored artefact rather than re-derived. A reader reproducing the analysis on different hardware should expect the same qualitative structure rather than an identical cluster count; pinning the thread count makes the result bit-for-bit portable.

The sub-family boundaries themselves are a separate question, and we tested them directly rather than leaving them provisional. Across 9 configurations of k and `n_neighbors` the recovered cluster count ranged from 2 to 6, and agreement with the adopted partition had a median ARI of 0.40 (worst case 0.04), an improvement on the superseded default (median ARI 0.21) but short of what a subfamily assignment would need.

The clusters do not correspond to distinct consensus sequences. A k-mer partition is not evidence of subfamily structure, so we cross-checked it against the consensus alignments (`run_cluster_crosscheck.py`), which is the test Andrew asked for. A consensus was built per cluster from 120 sampled copies, every sampled copy was projected onto every cluster consensus with `mafft -addfragments -keeplength`, and each copy was assigned to its nearest consensus by Kimura two-parameter divergence. If the clusters were subfamilies, copies would match their own consensus best. They do not. Of 599 copies only 36.1% are nearest their own cluster's consensus (chance 20.0%, worst cluster 7.5%), and the median copy is marginally *closer* to another cluster's consensus than to its own (divergence 0.035 versus 0.030, separation -0.005) (Figure 12). We therefore do not describe these clusters as subfamilies anywhere in this paper, and the phylogeny in Figure 6 should be read as a relationship among cluster consensuses rather than as a subfamily tree.

What the partition does capture is degradation state, and it captures it sharply: ORF integrity runs from 91.2% of copies intact in the most intact cluster to 0.1% in the least, and the expression reported in Section 2 concentrates in the intact-rich clusters. That is precisely the axis the workflow needs, because separating expressed, intact copies from inert relics (Section 2.2) is what makes the downstream reagents meaningful. The clusters are a useful stratification of copy state, not a reconstruction of ancestry, and we report them as such.

Scope of the reported results. The core workflow (Prep, Clustering, Alignment and consensus, Motif, GO, Expression, Primers) plus the phylogenetics, gRNA, transduction and antisense modules are the tested path and are the source of every number reported in Section 2. The modules in Section 6 are exploratory: they run, but they are not part of the validated result set for this manuscript.

5 Design and implementation

TERRA is organised in three layers (Figure 14). A desktop application built with Tauri and a React front end gives point-and-click access and performs no analysis itself; it launches a bundled Python sidecar and communicates over newline-delimited JSON on standard streams, so long-running jobs stream progress and can be cancelled. The analysis backend is a driver (`query.py`) that orchestrates the core stages, one module per stage, a family of standalone `run_*.py` modules, and an SSH client. In local mode everything runs on the user's machine, fetching sequence from the UCSC API [26] when no local FASTA is available. In HPC mode the code is staged to a login node, submitted as a batch job, and the results retrieved. Every external resource (RMSK/Dfam loci, UCSC sequence, JASPAR [24], mygene.info [27]) is reached through a thin adapter, so adding an assembly or swapping a source is a localised change.

The pipeline is a directed sequence of stages (Table 2). Each stage reads the previous stage's tabular output and writes a checkpoint, so a run can be resumed or a single stage re-executed. Figure 15 gives the end-to-end view.

Stage	**Function**	**Key methods / tools**
Prep	Loci, sequences, divergence	RMSK/Dfam [11]; FASTA or UCSC [26]; Kimura milliDiv
Clustering	Sub-group the copies	k-mers → SVD → UMAP [17] → HDBSCAN [18]
Alignment	Consensus per group	MAFFT [21] + CIAlign [28]
Motif	Enriched TFBS	JASPAR [24] ∩ loci (BEDTools [25]) + Fisher
GO	Gene/ontology context	mygene.info [27]
Expression	Per-group profiles + DE	per-locus counts (SQuIRE [3]); Wilcoxon + BH
Primers	Family/locus primers	k-mer candidates + genome-wide specificity
gRNA	Allele-aware CRISPRa/i	PAM-aware candidates + seed-anchored off-target scoring

Table 2: The TERRA core stages. Each stage consumes a tabular checkpoint and emits one, allowing partial re-runs. The gRNA and primer stages produce the deliverables of Section 2.5.

Engineering for the bottleneck. The dominant cost in whole-family primer design is reading the reference genome, since a naive implementation re-reads the multi-gigabyte FASTA for every candidate search. TERRA parses the reference once into an in-memory genome cache, reused for sequence extraction and all primer searches, and persists it so later runs skip parsing. A small number of batch operations (GC content, length tabulation, k-mer matrix construction) dominate the per-sequence arithmetic and have Cython [29] implementations with a pure-Python fallback.

Orchestration. The driver can run the copy-resolved modules as a sequential subprocess loop, or delegate them to a Nextflow DSL2 layer [6] that runs the same unmodified Python modules as parallel, individually-resourced, resumable tasks (Figure 16). No scientific code is duplicated between the two.

6 Exploratory modules

The following modules ship with TERRA and run as independent tasks, but are not part of the validated result set for this manuscript and are reported separately for that reason (Figure 17): CTCF-site and TAD-boundary proximity, epigenetic and regulatory track overlay, orthologous-insertion calling, multi-assembly liftover (`hg38`, T2T-CHM13, `mm10`, `mm39`), ColabFold structure prediction of consensus ORFs, pure-Python LTR structural classification, and automatic subfamily resolution into a dendrogram of cluster consensuses. Each emits its own figures and machine-readable values, and every run also writes a provenance manifest recording the git SHA, platform and tool versions alongside the stage checkpoints that enable resume.

7 Generality across TE classes

To establish that the workflow is not specific to a LINE, we applied the identical pipeline to a SINE and an LTR family in `mm10` with the same six-stage SQuIRE dataset. *B1_Mus2* (SINE) has 70,685 loci with 21,407 carrying expression and a median Kimura divergence of 7.5%. *IAPLTR1_Mm*

(LTR) has 1,485 loci with 1,426 carrying expression and a median divergence of 1.6%. The divergence medians order the three families by evolutionary age, with *IAPLTR1_Mm* youngest and most homogeneous, *L1Md_T* intermediate at 3.5%, and *B1_Mus2* oldest and most diverged, which is consistent with their known retrotransposition histories [1]. Full per-family reports for all three families, including the clustering, expression, alignment, consensus, motif, primer and guide outputs, are provided as Supplementary Files.

8 Discussion

TERRA fills a specific gap: the accessible, reproducible, cluster-ready path from a named TE family to reagents that can be ordered. It complements rather than replaces discovery tools such as RepeatModeler2 [2], expression quantifiers such as SQuIRE [3] and TEtranscripts [4], and general platforms such as Galaxy [14] and nf-core [7].

Its contributions are threefold. First, a conceptual one: resolving a family into sub-groups and intersecting them with locus-level expression, rather than summarising the family as a unit, is what allows a signal carried by a minority of copies to survive the analysis, and it motivates the distinction between the expressed and the common consensus (Section 2.2). Second, a practical one: treating primers and allele-aware CRISPRa/i guides as the deliverable, and reporting the guide coverage/off-target trade-off explicitly as a frontier rather than collapsing it into one score (Section 2.5). Third, an engineering one: the in-memory genome cache, the compiled hot paths, and automatic LSF/Slurm support make whole-family analysis practical without rebuilding the workflow when moving from a laptop to a cluster.

Several limitations bound the system. Expression analysis requires user-supplied per-locus counts, so quantification quality and its multi-mapping policy are inherited from the upstream tool. Family resolution and sequence depend on external services (RMSK/Dfam, UCSC, JASPAR, mygene.info), so a run is only as reproducible as the database releases it draws on, and supported assemblies are currently the common human and mouse builds. The k-mer clustering, while fast and alignment-free, does not yield stable sub-family boundaries at the default minimum cluster size: agreement between parameter settings is close to chance (median ARI 0.40, Section 4), so we report the partition as a device for separating expressed from inert copies rather than as a subfamily assignment. Resolving this requires the parameter search described in Section 4 followed by cross-validation against the consensus alignments. The guide frontier likewise reports sequence-level predictions that require experimental confirmation. The modules in Section 6 are not yet validated to the standard of the results reported here.

Natural extensions follow from the architecture: per-locus epigenomic context (WGBS, ATAC-seq) to separate silenced from derepressed copies; correlating per-cluster TE activity with nearby gene expression to nominate regulatory TEs; somatic-insertion calling from user BAMs; broader assembly support; and negative-binomial modelling of count overdispersion using established frameworks [30, 31].

Code availability

TERRA is open source at `https://github.com/anmol-dash/te-scraper-and-analysis`. Figures are generated by the pipeline scripts in that repository, which run the analysis locally or on an HPC cluster and emit the figures and measured values cited here.

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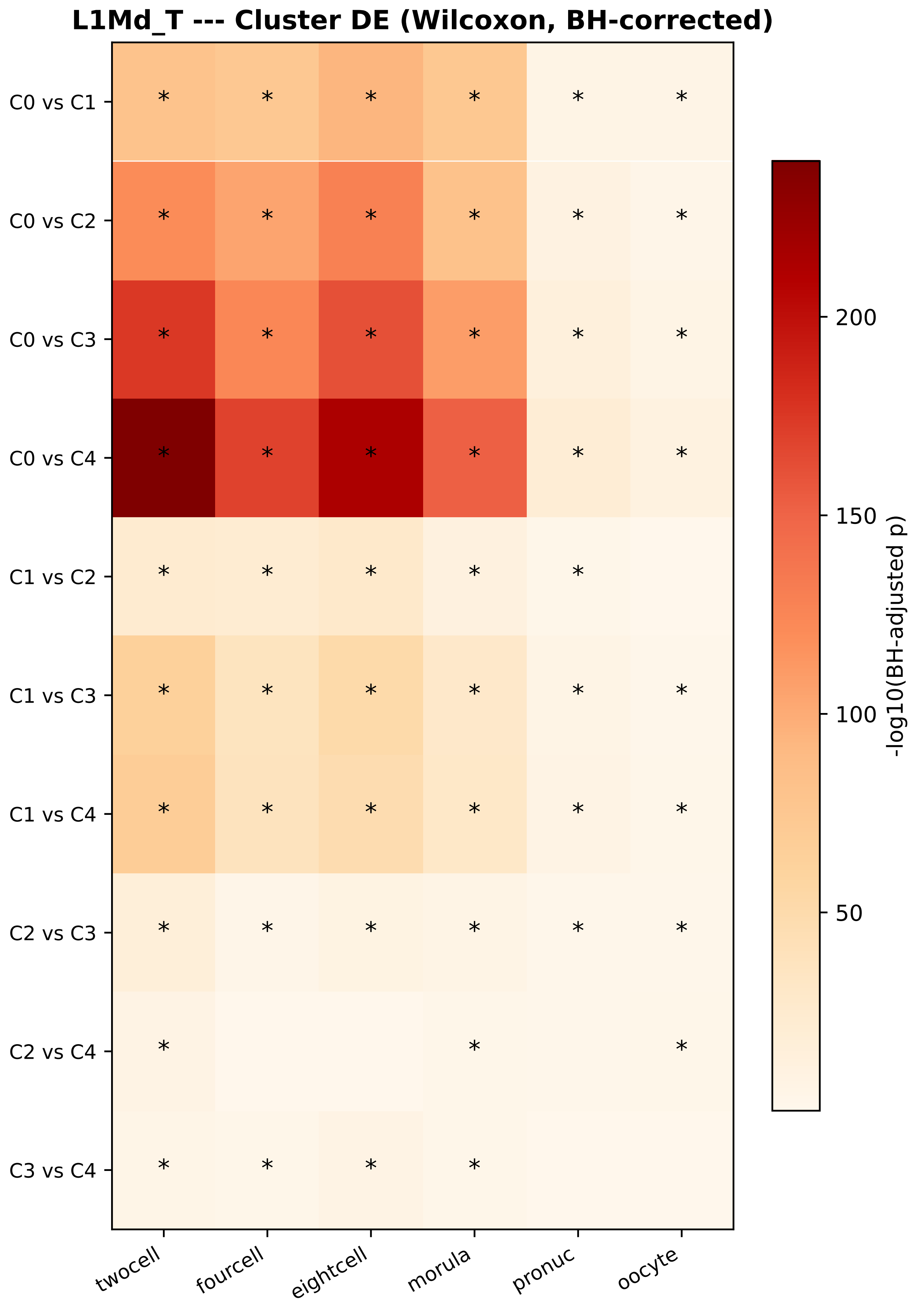

Figure 5: Differential expression between *L1Md_T* clusters. Heatmap of $-\log_{10}(q)$ from pairwise Wilcoxon rank-sum tests with Benjamini–Hochberg correction; 54 of 60 cluster-pair by stage comparisons reach $q < 0.05$. Cluster-resolved testing recovers stage-specific differences that a family-level summary averages away (Section 2.2).

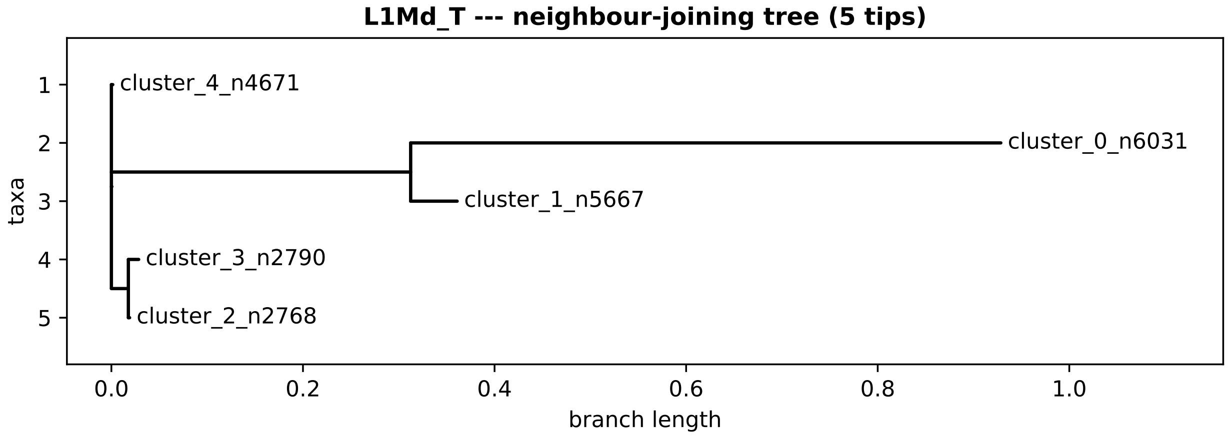

Figure 6: Subfamily phylogeny for *L1Md_T*. Neighbour-joining tree relating cluster consensuses and representative copies, built from the MAFFT alignment.

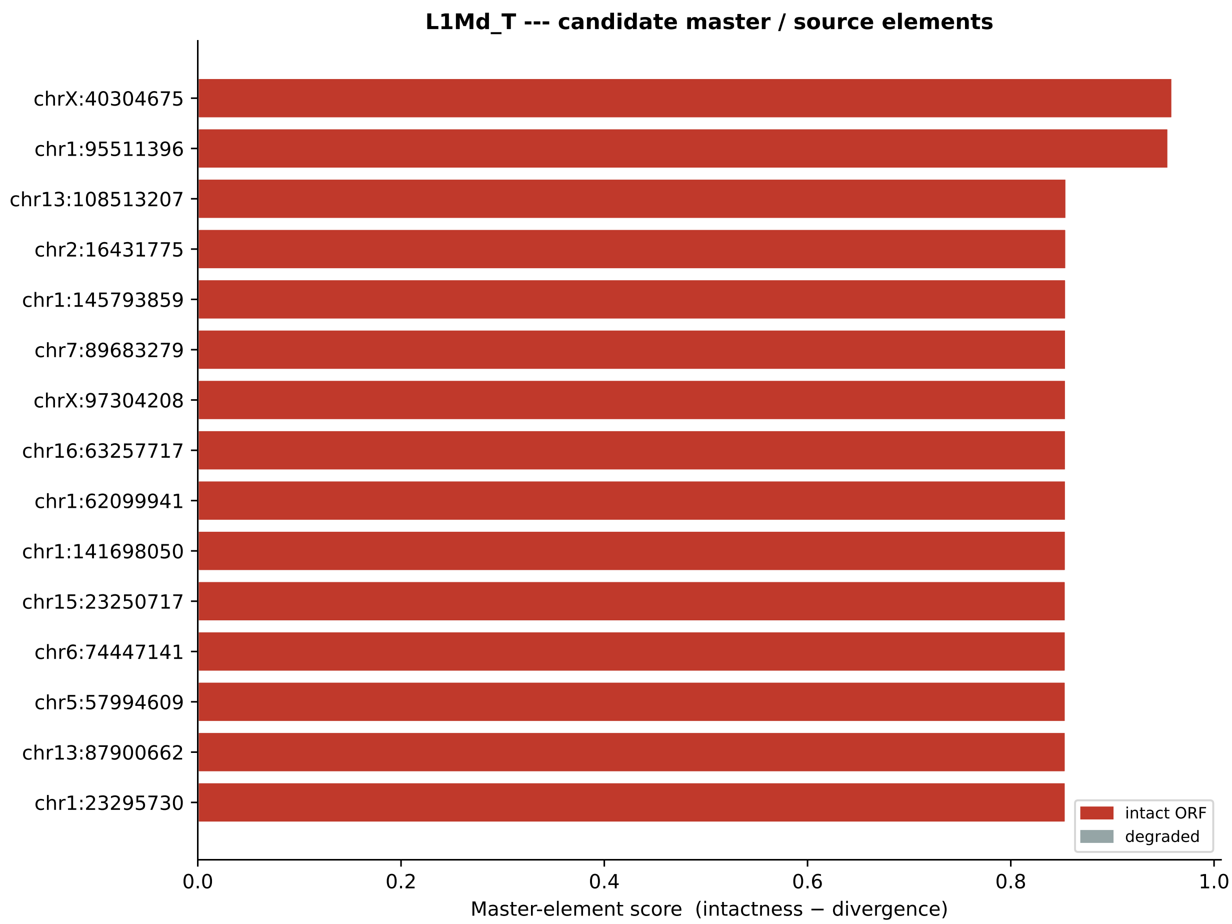

Figure 7: Master-element nomination for *L1Md_T*. Copies are scored on ORF integrity against divergence from the consensus; 9388 copies are intact and the top-ranked candidate is `chrX:40304675`.

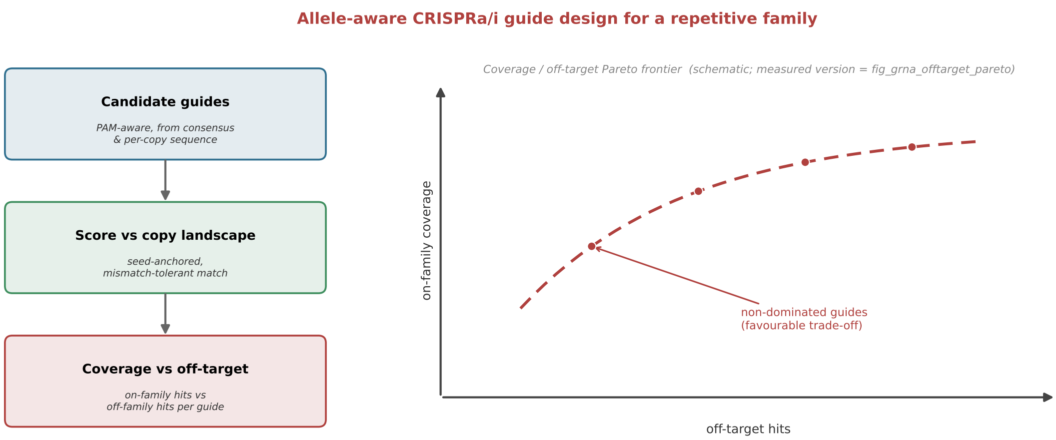

Figure 8: Allele-aware CRISPRa/i guide design for a repetitive family. Candidate guides are enumerated PAM-aware from the consensus and per-copy sequence, scored against the full copy landscape with seed-anchored, mismatch-tolerant matching, and reported as a coverage-versus-off-target trade-off. This panel is a schematic of the logic; the measured frontier for *L1Md_T* is Figure 9.

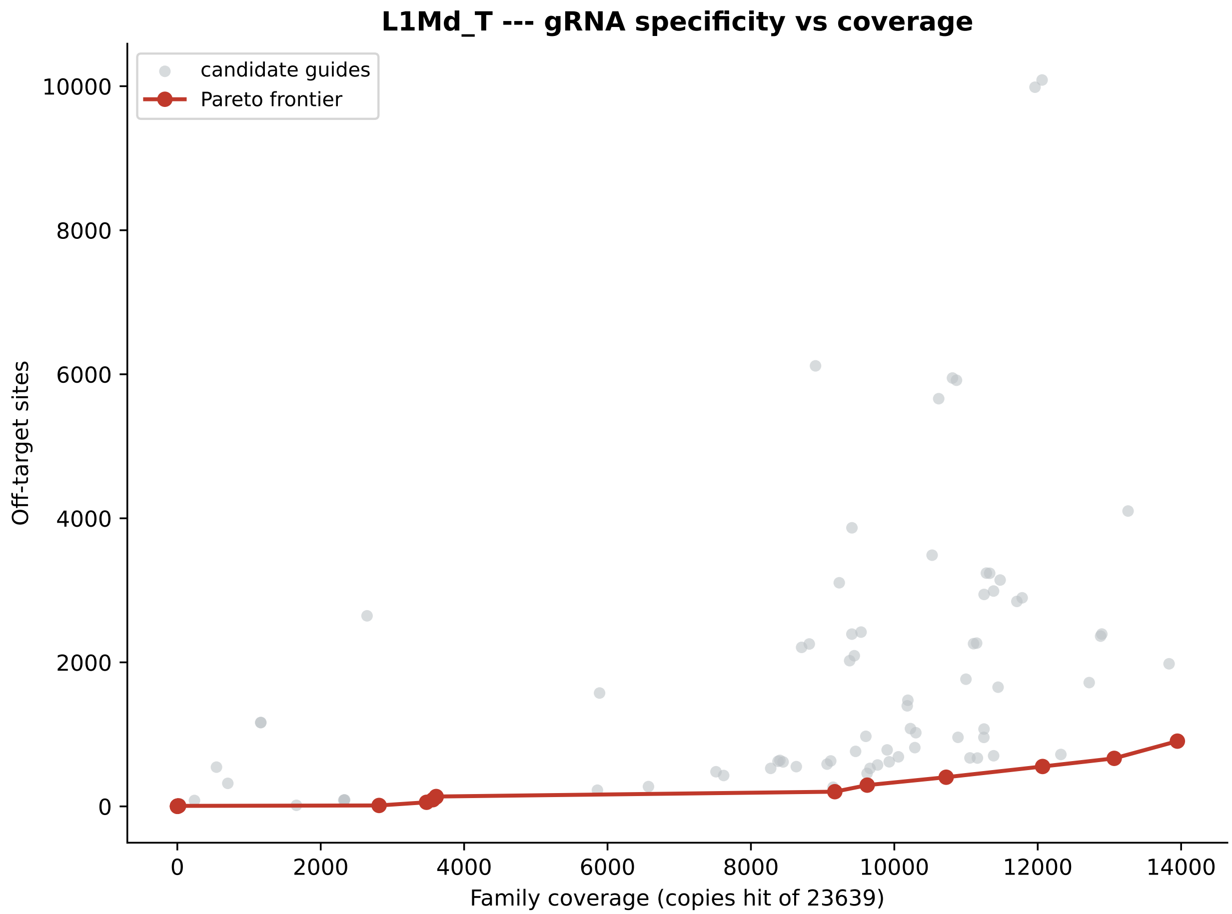

Figure 9: Measured coverage-versus-off-target frontier for *L1Md_T*. 400 candidate guides scored against 23,639 copies; 17 guides are non-dominated. Each frontier point is a defensible design choice at a stated coverage and off-target count.

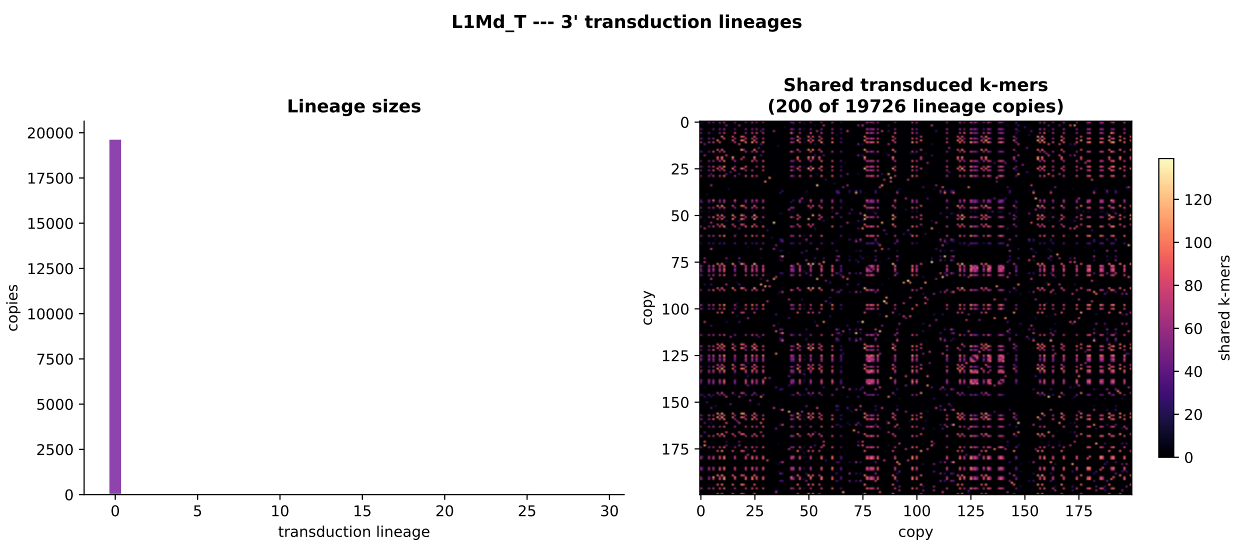

Figure 10: 3' transduction lineages for *L1Md_T*: 30 candidate lineages grouped by shared downstream-tail k-mers absent from the family consensus; 8227 copies carry an AATAAA poly-A signal.

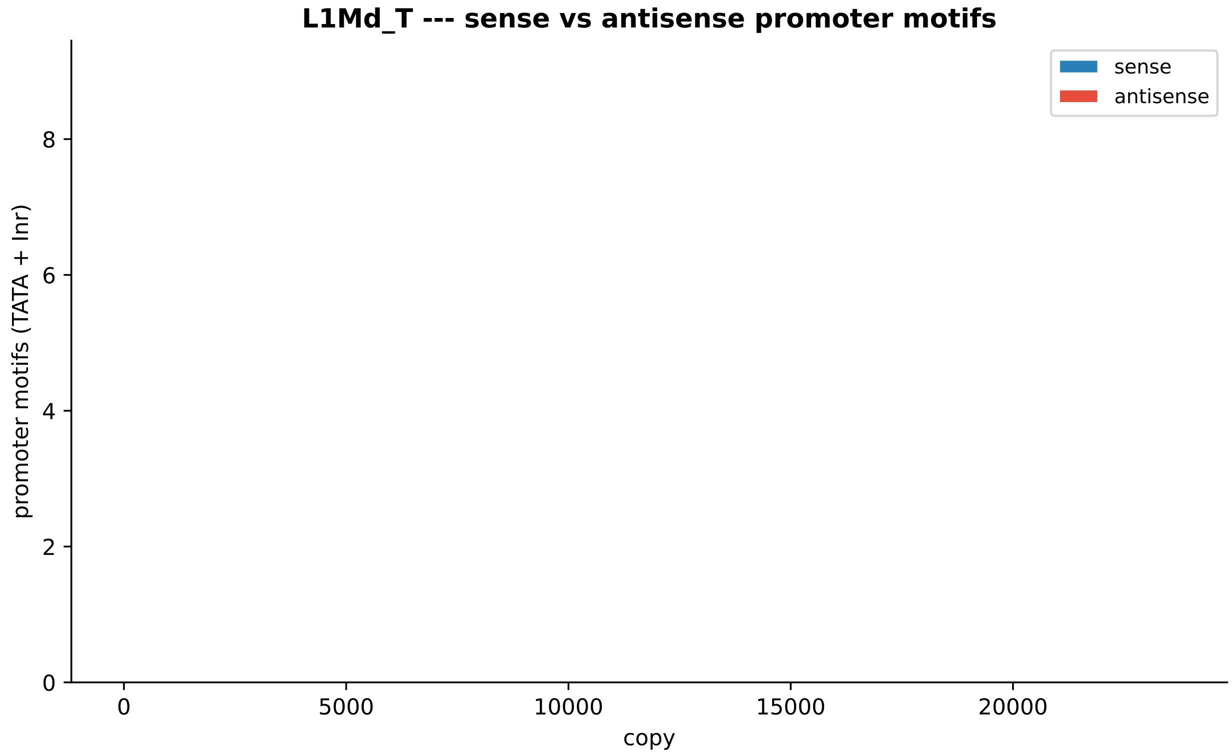

Figure 11: Antisense/bidirectional promoter content for *L1Md_T*: core promoter elements scanned on both strands of each 5' region; 13246 of 23,639 copies are flagged bidirectional.

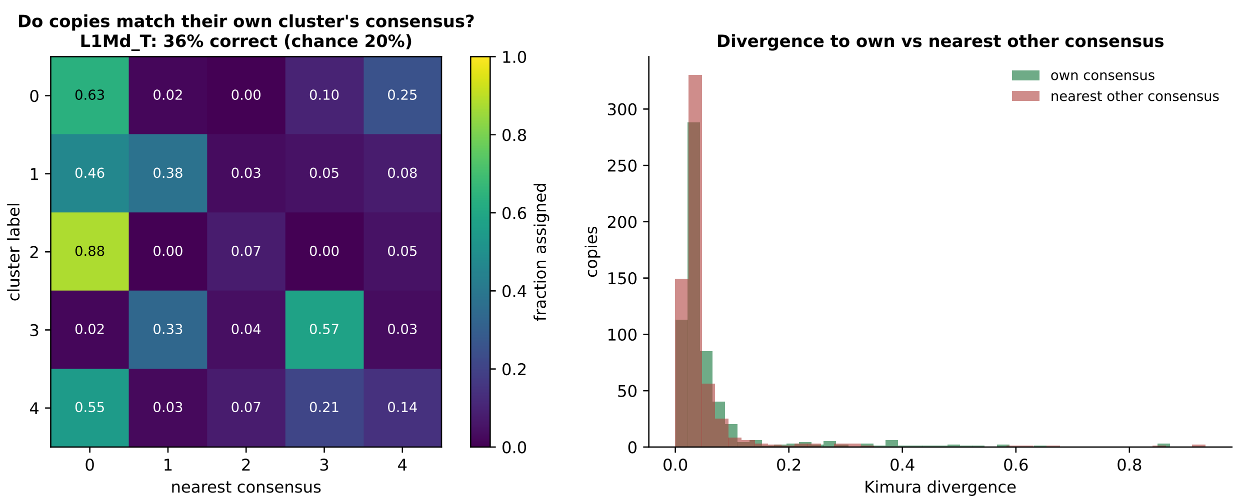

Figure 12: Cross-check of the clusters against the consensus alignments for *L1Md_T*. **Left:** fraction of each cluster's copies whose nearest consensus is each cluster's, after projecting every copy onto every consensus with `mafft -addfragments -keeplength`. A subfamily assignment would put the mass on the diagonal; the observed 36.1% (chance 20.0%) does not. **Right:** Kimura divergence of each copy to its own cluster's consensus versus to the nearest other consensus; the distributions overlap, with the median copy marginally closer to another cluster's consensus.

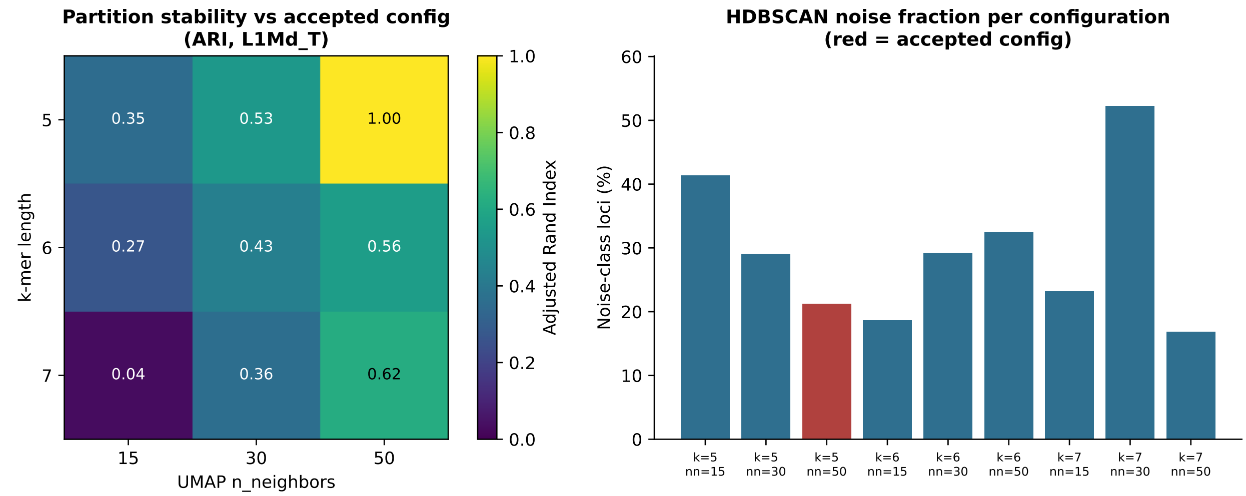

Figure 13: Clustering and noise-class validation for *L1Md_T*. **Left:** Adjusted Rand Index of each k and `n_neighbors` configuration against the configuration the automatic search accepted. Values near 1 would indicate the partition is independent of the parameter choice; the observed median of 0.40 indicates it is not, which is why these sub-family boundaries are not interpreted as biological. **Right:** HDBSCAN noise-class fraction per configuration, with the accepted configuration highlighted. Noise-class loci are retained as a labelled group and excluded from consensus building.

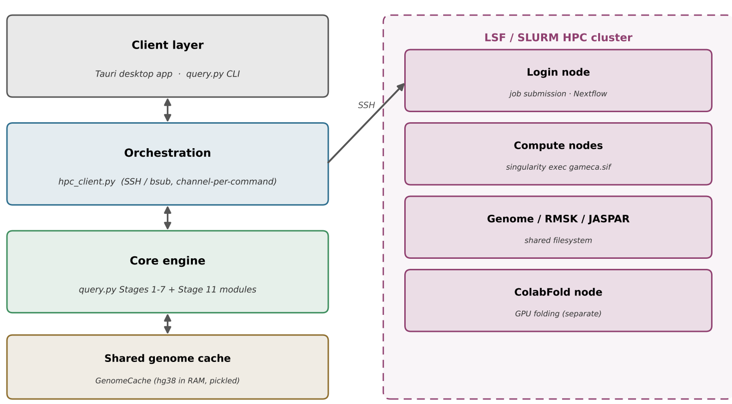

Figure 14: TERRA's three-layer architecture. A Tauri/React desktop shell drives a Python analysis sidecar over a streaming NDJSON protocol; the backend runs locally or stages and submits jobs to an LSF/Slurm cluster. Coordinate, sequence, motif and annotation data are pulled from RMSK/Dfam, UCSC, JASPAR and mygene.info through uniform adapters.

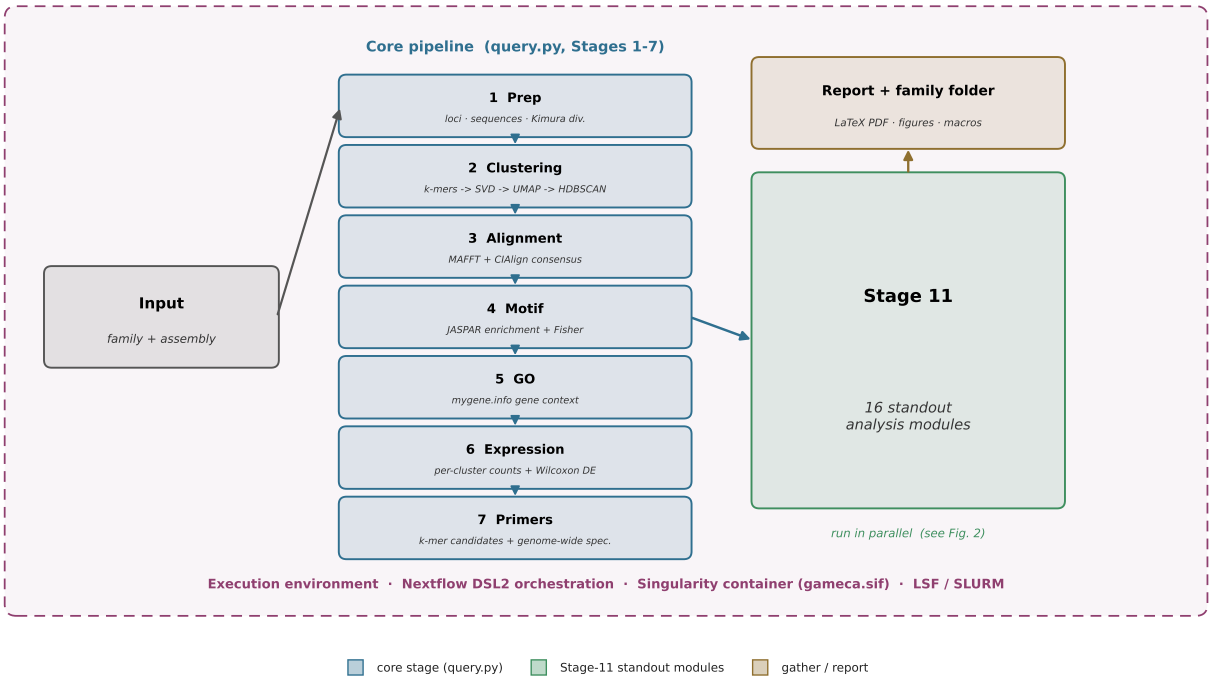

Figure 15: TERRA pipeline flowchart. A family/assembly request flows through the core stages and into the copy-resolved modules, which run in parallel; outputs are gathered into a per-family folder and report. The run executes inside the Nextflow DSL2 and Singularity environment on an LSF or Slurm scheduler.

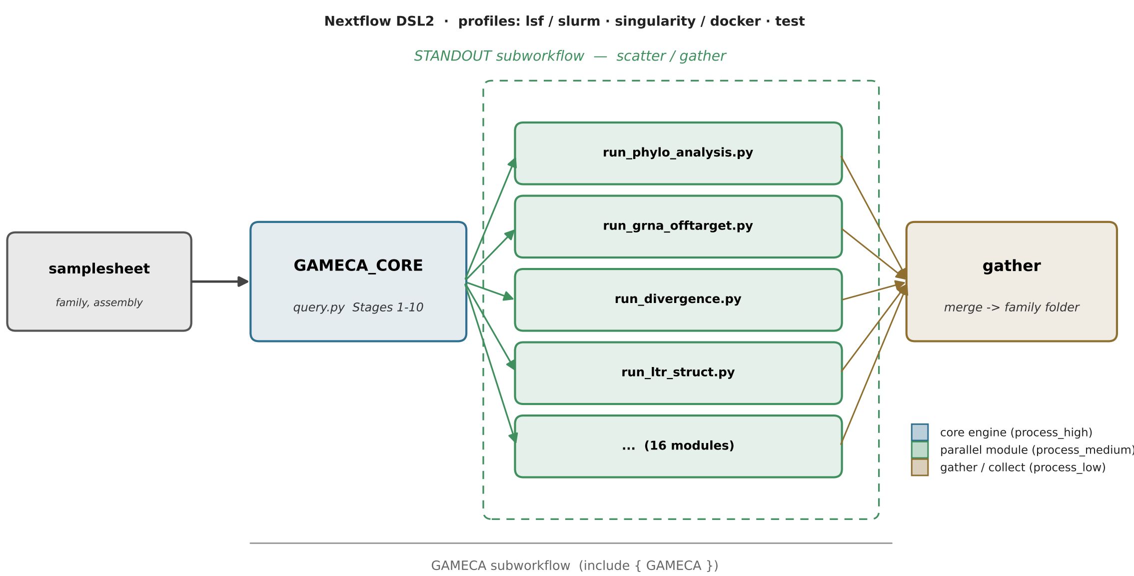

Figure 16: The Nextflow DSL2 execution graph: the core engine feeds a scatter of independent copy-resolved module tasks, gathered into a single per-family output. Profiles select LSF or Slurm and Singularity or Docker.

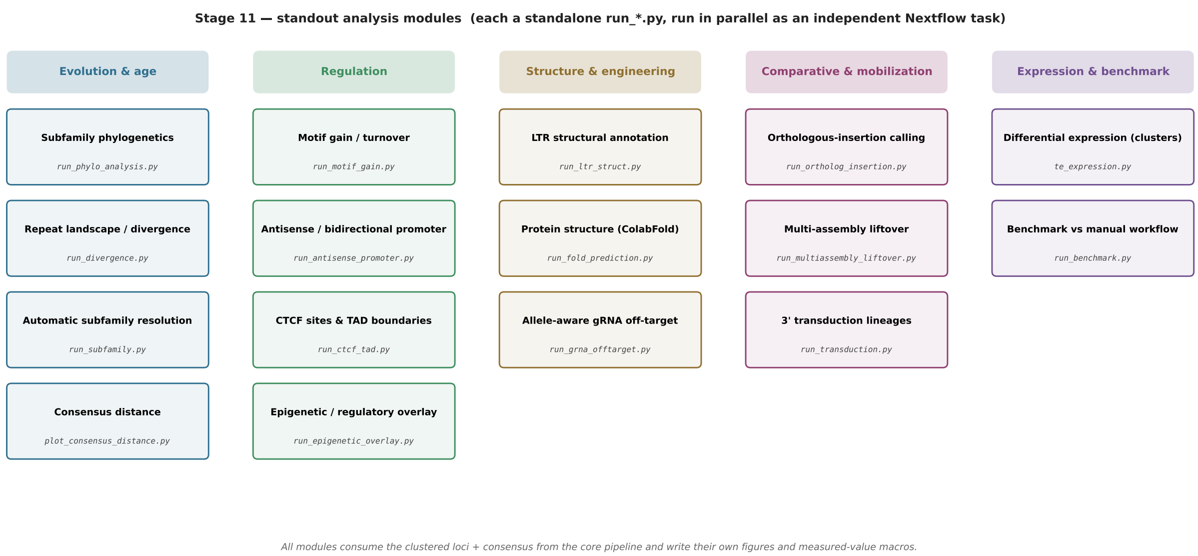

Figure 17: The copy-resolved analysis modules grouped by biological theme. Each card names the standalone `run_*.py` script. The phylogenetics, gRNA, transduction and antisense modules are reported as results in Section 2; the remainder are exploratory (Section 6).